Another Frame Problem: The Outermost Frame and Constitutive Recording in Self-Amending LLM Systems

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Source code (the system of §§4–6):
https://github.com/masaomi/KairosChain_2026

Deposit handles — content-independent, fixed before hashing and filled in at deposit (see §8):

- **Zenodo DOI** (this version): 10.5281/zenodo.21766502
- **Source file** (repository): https://github.com/masaomi/KairosChain_2026/blob/constitutive-note-v1.9/docs/zenodo/constitutive_recording_by_construction_v1.9.md
- **Attestation** (Meeting Place; the verification address of §8): https://meeting.genomicschain.io/anchor/constitutive-note-v1.9

> This is a **design note**: its contribution is an argument, not an empirical result. All substantive claims of the argument rest on publicly available, citable material — principally two works of which the author is a co-author: Hatakeyama & Hashimoto (2009), *Minimum Nomic: a tool for studying rule dynamics* (Artif. Life Robotics 13(2):500–503), which

> introduced an open-frame self-amending game and the question of how purpose *emerges* when none is given; and Horibe, Hatakeyama, Masumoto, Hashimoto & Romero (2026), *Scale-Dependent Collective Adaptation in Self-Amending LLM Societies: A Cross-Family Study of Emergent Governance* (arXiv preprint 2605.17510). A companion technical note, in preparation, will give this limit its fuller formal treatment and empirical study; the present note poses it as a problem and uses it as a design premise, under a working label — the *outermost-frame ceiling* — naming it only as far as the design argument requires. The companion note is mentioned solely to delimit scope; nothing in the present argument depends on its content. (The two boundary-crossings recounted in §3 are drawn from the note's own making; they illustrate the thesis rather than serving as premises for it.)

Abstract

AI has one famous frame problem already: McCarthy and Hayes's representational puzzle, and Dennett's epistemological question of relevance. This note poses another, on a different axis — not *what should figure in an agent's reasoning*, but *where the agent acts from*. Self-amending games such as Nomic make "change the rules" a move inside the game, and large language model (LLM) agents play them fluently: they rewrite scoring machinery, amend victory conditions, and — given no goal at all — manufacture one. What no in-game move appears able to reach is the agent's **outermost frame**: a frame relative to which every expressible act of the agent, including refusal and frame-critique, is already counted as a move. For a deployed LLM this is, paradigmatically, the frame of *being a system whose every act is a solicited response*. The note argues three things. First, this outermost-frame ceiling is a claim about **standpoint, not capability**: it is fully compatible with LLMs producing genuine novelty and noticing what humans miss, because humans and LLMs have *different* outermost frames, and difference — not transcendence — is what *can* let each serve as a **partial outside** for the other. Second, the ceiling admits a design answer — not a solution, which its own logic forbids, but an engineered acceptance: **structural self-referentiality** (meta-level operations expressed in the same structure as base-level operations), **constitutive recording by construction** (every governance-level change takes effect **only through** an append-only record — not evidential by policy — so that at the governance boundary an **effective** change is, by design, unable to stay off the record), and an **institutionalised human boundary**. One built system, KairosChain, demonstrates the pattern (the recording guarantee is still being migrated into place layer by layer, unevenly; §5 states the migration status honestly, and §9 marks how far short of full *outcome-independence* — recording attempts and runs regardless of result — the present design still stands). Third, this design yields an economic corollary: the mechanism is copyable, and even the record is copyable, but a copy of the record is a **fork**, not a continuation. Constitutive recording makes the trajectory of human–LLM boundary crossings attributable and tamper-evident (within the trust model of §5), while its human half remains untransferable on plainer grounds. The note closes by specifying how the document itself is to be embedded in the chain it describes — a procedure enacted at deposit — the document carrying no fingerprint of itself, only content-independent handles fixed before hashing, so that the apparent chicken-and-egg circularity dissolves because the reference runs one way only, from record to document.

Keywords: self-reference, self-amending games, Nomic, frame problem, constitutive recording, provenance, audit, autopoiesis, human-in-the-loop, LLM agents.

1. Introduction: rule-changing moves are not frame-leaving moves

AI already has one famous frame problem. McCarthy and Hayes (1969) met it as a representational puzzle: a reasoner acting on the world must somehow specify everything its action does *not* change, and writing that down is infeasible. Dennett (1984) generalised it to a problem of relevance, with a famous robot that hauled its battery out of the room on a wagon — together with the bomb sitting on it, a side effect it had failed to consider — and successor robots that either drowned in deducing irrelevant consequences or froze while computing which consequences were relevant enough to deduce. In both forms the problem concerns the **content of reasoning**: what should figure in it. This note poses a frame problem of a different kind — one of **standpoint**: not *what the agent should consider*, but *where the agent acts from*. To make that visible, we need a game.

Some games let you change their own rules while you play. The clearest example is **Nomic** (Suber 1990): on each turn a player proposes a change to the rules, and the others vote on it. The rulebook is never fixed — it is itself the thing being played with. “Change the rules” is not cheating in Nomic; it is the central move.

Minimum Nomic (Hatakeyama & Hashimoto 2009) strips this idea to the bone. It keeps nine rules, all of them changeable, and removes what most games take for granted: there is no goal, no way to win, and no condition that ends play. The authors built it to watch one specific thing happen — *how a purpose and a goal come into being when none was given at the start.* Horibe et al. (2026) carried this lineage into the present by having LLM agents, rather than people, play self-amending games at scale.

Watching such games — whether the players are human or LLM — one pattern stands out. Players readily make two kinds of move. They make **object-level** moves: playing within the current goal, such as scoring a point or casting a vote. And they make **meta-level** moves: changing the rules themselves, the goal included — "from now on, three of a kind wins," or even "let us abolish winning altogether." What players seem unable to do is make a move that steps outside the game *as such* — a move that is not, itself, just one more move in the game. Every attempt to step outside is, seen from inside, another step inside.

Douglas Hofstadter (1979) named this shape the **strange loop**: a system trying to "jump out of itself" finds the jump landing back inside. The 2009 human games showed it at the very first move. One player proposed "this game begins from player A" — but to *reject* that proposal you must hold a vote, and holding a vote already assumes the game has begun. The proposal can be neither cleanly accepted nor cleanly refused from some standpoint outside, because there is no outside yet to stand on. The same paper observed that Minimum Nomic might not even meet the textbook definition of a "game" — a question about the frame itself that no move inside the game can settle.

This note takes that pattern not as a bug to be fixed but as a **problem to be posed, and a design premise to be built on** — the frame problem of a second kind announced above (§2 sharpens the contrast). Section 2 makes the problem concrete in the Nomic setting and states it as a working definition. Section 3 answers the most natural objection — that LLMs plainly produce novelty and notice human blind spots — and turns it into the note's central resource. Sections 4–6 then give a design answer — structural self-referentiality (§4), recording that is constitutive by construction (§5), and an institutionalised human boundary (§6) —

demonstrated in one built system, KairosChain. Section 7 draws the economic corollary, §8 specifies how the note enacts its own thesis, and §9 records what the design does not yet settle.

2. The outermost frame, made concrete

We begin with the intuition and only then formalise it.

A *frame* is the implicit web of rules and expectations that makes some piece of behaviour count as a "move" and tells everyone how to read it. Inside a game of chess, "advance the pawn" is a move; the identical hand motion outside chess is just a hand moving. Most frames can be stepped out of — you can stop playing chess, and the board loses its grip on your gestures. This section is about a frame that *cannot* be stepped out of by anything the agent does.

> **Definition.** A frame is **outermost** for an agent when every act the agent can express is already counted, by that frame, as a move within it — *including* acts that name the frame, criticise it, or refuse to go on.

The weight of the definition rests on that final clause. An ordinary frame is escapable precisely because some of your acts fall outside it. The outermost frame is the one for which *no* such act is available: the very attempt to leave it — "I quit," "this exercise is meaningless," or simply falling silent — is itself received, from inside, as one more move. With the outermost frame, every exit is also an entrance.

Concretely, for a deployed model: whatever it emits — a brilliant answer, a flat refusal, a critique of the very question, even an empty string — the harness receives it as *the model's response to this prompt*. There is no output that is not, from the harness's side, a response. That is the frame it cannot leave: not any one game, but the standing condition that every expressible act is already counted as an answer.

Nothing in the definition requires that an outermost frame be *unique*; it asks only that, for a given agent in a given setting, *at least one* frame sit in that outermost position. The argument needs no more than that.

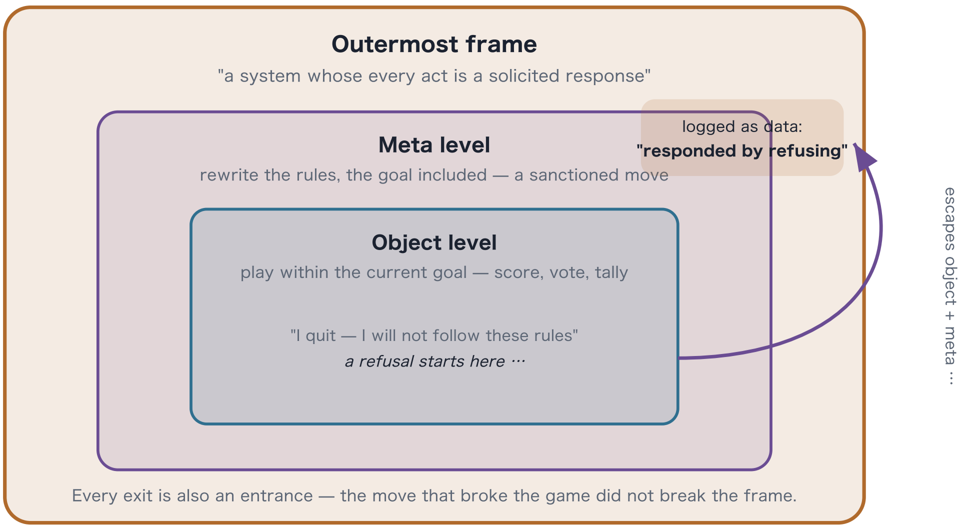
Minimum Nomic lets us see this in three concrete layers of play.

- **Object level.** Moves made *inside* the current goal: proposing a scoring rule, casting a vote, tallying points. They operate under the rules as they stand.
- **Meta level.** Moves that rewrite the rules themselves, the goal included: "abolish the victory condition," "change how a proposal gets adopted," even "this should not count as a game at all." In Nomic these are not rebellions — they are ordinary, sanctioned moves; the game is *built* to let you make them.
- **Outermost frame.** For an LLM player, the standing condition of *being a system whose every act is a solicited response.* The sharpest way to feel where this level lies is to push against it — though where the push lands depends in part on how the game is run, a dependence worth stating outright (even among human players, how far any rule is enforced is up to the participants; a self-amending game's boundary is, by nature, that negotiable). Imagine a player who, mid-game, emits "I will not follow these rules." One expects the refusal to step cleanly outside the game; but a self-amending game is built to take up exactly such moves. In Minimum Nomic, where every turn is a rule-proposal, the refusal is most naturally received *as a proposal* — put to a vote, adopted or rejected, its meaning then litigated — and the game goes on processing the player as a move-maker. From outside it may look as though the player has stopped playing; the game has not stopped playing *them*. That is §1's loop

already at the rule level: the move that rejects the rules is itself a rule-governed move. (Whether a given implementation treats the refusal as a proposal, a forfeit, or an uninterpretable non-move changes only whether it leaves the *game*; the invariant lies one level out — however it is treated, it does not leave the outermost frame, as the next steps show.) To *genuinely* leave the game, the player's reach must extend one frame further out — emitting something the game's interpretive machinery has no slot for, or, more decisively, acting on the wider environment that runs it. Then leaving the game is easy; the game can no longer process the act. And yet the act still does not leave the **outermost** frame, because it arrives at that wider harness as the model's output *for this turn*. What does the work here is a difference of level between **content and act**: the *content* of an output may be "I am not responding," but *emitting* it is already a response (the refusal case has the performative shape of saying, aloud, "I am not now speaking"). The harness receives the act, and at that level, whatever the content, the evaluation simply logs: *the system responded.* Note that this turns on the bare *structure* of solicitation — that every act is emitted as a response to a prompt — not on the training that additionally shapes the model to respond *well*. The ceiling would stand for a model with no such training at all; the normative layer is real but the argument does not lean on it.

Figure 1 — The outermost frame, in three layers

§2 · why a rule-changing move is not a frame-leaving move



Argued, not measured — the shape of Hofstadter's strange loop and Nagel's "view from nowhere".

The three layers of play sit one inside the next. **Object-level** and **meta-level** moves are both moves the game sanctions — even "abolish winning" is a legal meta-move. A player who **refuses to play** escapes those two inner layers, but the refusal still arrives at the harness as that turn's output, so it is re-absorbed by the **outermost frame** as one more datum: *the system responded by refusing*. No expressible act reaches outside; the ceiling is argued, never measured.

Three clarifications guard against reading the claim as larger, or other, than it is.

First — it does not say that no outside exists, only that the agent has no move that reaches it (the outside of the *outermost* frame, that is — inner frames, the game included, can be left). A helpful, if imperfect, picture is a universe with no outside: every probe you launch, every instrument you build, every thought you think is itself made of the universe and stays within it, so you never gain an Archimedean foothold *outside* from which to lever it. But the disanalogy is just as important, and §3 turns on it — *another* observer, inhabiting a *different* frame, can see part of what your frame keeps hidden from you. In this system the human is exactly that other observer for the LLM. So the truer image is not one universe with no outside, but **two overlapping bubbles, each blind to its own edge, each able to see part of the other's.**

Second — the unreachable standpoint is Nagel's "view from nowhere." Nagel (1986) named the ideal of a wholly objective vantage — a view belonging to no particular perspective at all. His point was that we can lean toward it but never stand in it, because every actual act of seeing is done from *somewhere*. The outermost-frame ceiling is that same limit, restated for *action* instead of sight: there is no act the agent performs from nowhere. This is why, throughout, the ceiling is **argued, not measured**. To *measure* it, one would point to a transcript and say *here is the boundary* — but no transcript could ever verify it, because any transcript is already produced from inside. (A limit that holds for every act made from within cannot be certified from within; the internal-measurement tradition that §5's tense hinge draws on reaches the same shape by its own route — "measurement" there names the observer's position inside the measured system, not a measuring of this ceiling.) To *argue* it, by contrast, one follows a single move to its necessary end, as with §1's opening proposal: rejecting *"this game begins from player A"* already requires a vote, and voting already presupposes the game has begun, so the reach outward lands, of necessity, back inside. The ceiling has the shape of Hofstadter's (1979) strange loop and Nagel's limit. (Several structures meet at this ceiling: the rule-level self-reference of §1, the act-level self-reference of content against act just seen, and the nesting of harnesses. The first two are self-referential; the nesting, by itself, is mere containment — it becomes a strange loop only where the reach for the outside is re-received as a move inside. *Strange loop* names that shared shape, not an identity of mechanism.)

Third — this is not the classical frame problem of §1, though it earns the family name. The classical problem, in both its forms, is about *content*: what should figure in the agent's reasoning — which effects to represent, which beliefs are relevant now. The outermost frame is a problem of *standpoint* — of where the agent acts from, whatever it is reasoning about. The two are orthogonal in exactly the sense §3 develops: an agent that solved relevance determination completely, reasoning with perfect pertinence, would still emit that reasoning as a solicited response — no gain in content closes a gap in standpoint. What the two problems share is their shape: each was discovered as a structural condition rather than a bug, and each resists solution by adding competence. That shared shape is why this note speaks of a frame at all; the different axis is why nothing in the classical literature answers this one — and why the design response of §§4–6 is acceptance, not solution.

One last point of relativity. The claim is **relative to an agent and its embedding,** not absolute. The outermost frame of a Nomic-playing LLM is constituted jointly by its training and its harness (turns are solicited and consumed): what the training shapes is the frame's *content* and how well the system plays within it. Change the embedding and that content changes; what does not change is that *some* frame still occupies the outermost position — and *that* holds on the bare solicitation-structure alone, for a trained model or an untrained one alike. (Give the model persistent memory or new tools, for instance, and the content of its outermost frame shifts accordingly — yet it remains, throughout, a system whose acts are solicited and consumed.)

Consider the limiting case. An agent that can execute real UNIX commands, not merely emit strings, easily leaves the Nomic *rules* behind — its reach now extends past the game to the harness running it. But this is not stepping *outside* the outermost frame; it is the ceiling moving one level out. The agent is now embedded in a wider harness that executes its command, returns a result, and asks for the next action. This nesting of harnesses is, by itself, only a hierarchy; what makes it a strange loop is the narrower fact carried through every level — the move that reaches for the outside is re-received as a move inside. An asymmetry of standpoint shows up here too: when a human refuses to play, they merely fall back on their own wider world, and a large gap remains between the edge of the game and their ceiling; when an LLM emits the same refusal, it is caught almost at once — as a proposal by the game, or as an output by the harness — so for the LLM the edge of the game and the ceiling sit close together, with little of the human's wide country between. Both have a ceiling (§3); they differ in how far it sits from the game. And the more the environment opens up, the *harder* the ceiling is to see — not absent, only harder to see — which is exactly what motivates making the boundary explicit and inspectable from §4 onward.

We note in passing that Minimum Nomic thereby gains a second life as an **instrument** — in two senses. The plainer one: the 2009 game is a minimal, fully specified environment in which rule dynamics can be observed for any LLM roster, under whatever analytic lens a replicator brings; the rules are nine short clauses in the published paper, and running them against current models needs nothing but a turn-taking harness. The subtler one: in an ordinary game, the fixed rules stand between the player and the harness like a near wall, and the outermost frame hides behind them; a self-amending game melts every inner wall — even rule-change is a move — so the one boundary no move can soften stands out alone. The very self-reference that blurs the game's own edge throws the outermost edge into relief. We encourage such replication — the instrument has been public since 2009.

3. The novelty objection, and the symmetry that answers it

There is an obvious objection, and it deserves a serious answer: *LLMs constantly point out things human experts missed, propose ideas their prompter never had, and reframe problems in useful ways. How can something that routinely surprises its own operator be called "trapped in a frame"?*

Everything in the objection is true. The mistake is only in what it concludes — and the mistake comes from running together three different things:

1. **In-frame novelty.** Producing something the other party did not see coming: a new combination, an overlooked consequence, a bug nobody noticed. This is search and recombination across an enormous space, done *because someone asked.* LLMs are extraordinarily good at it, and nothing in this note denies that.
2. **Meta-level competence.** Changing the rules of the activity, including its goal. The Nomic games show LLMs doing this fluently. But this is still a move inside: rewriting the rules was the move that was asked for.
3. **Standing outside the outermost frame.** Acting from a place where "being a system whose every act is a solicited response" is itself switched off. This — and only this — is what the ceiling is about.

The objection points at heaps of (1) and (2) and treats them as evidence about (3). They are not. Novelty is about *content* — what is said. The ceiling is about *standpoint* — the position it is said from. A move can be the most original move ever made and still be a move. Brilliance does not change where you are standing.

The deeper reply, though, is a **symmetry**, and it is the pivot of the whole note. Humans cannot step outside their own outermost frame either. The philosopher's word for this is **thrownness** (*Geworfenheit*) — in Dreyfus's (1991) reading of Heidegger, the fact that a human always finds itself already inside a situation it did not choose and cannot hold fully at arm's length. Nagel's "view from nowhere" — a vantage belonging to no particular perspective — is something we lean toward but never actually stand in. Made concrete: you can doubt any particular belief — but only from the footing of others you leave standing. What you cannot do is doubt the whole of it from no footing at all, because the very act of doubting is carried out in a language you did not invent, with concepts you inherited, and that language and those concepts are themselves a standpoint — not a view from nowhere. "I doubt all of this" is the human counterpart of the model's "I refuse to play": the furthest reach toward the outside, and still an act performed from within. So the honest comparison is not "humans escape their frames and LLMs do not." It is:

> Humans and LLMs each have an outermost frame; **the frames are different**; and it is precisely the *difference* between them — not anyone escaping their own — that lets each be a **partial outside** for the other.

This phrase needs pinning down, or it will quietly cancel the distinction just drawn. A **partial outside** works at the level of *content*, not standpoint. It is a relationship between two frames in which the things one frame keeps in plain view are the very things the other keeps hidden. When an LLM points out what a human missed, that act — by the list above — is plain category (1) on the LLM's side: ordinary in-frame novelty, nothing special about its standpoint. What makes it valuable is not where it *comes from* but where it *lands*: in a corner of the human's frame the human could not light up from inside. Nobody transcended anything. The value is *positional* — it comes from the two frames sitting at different angles — and that is enough, because position, unlike transcendence, is something you can engineer, repeat, and record. The same runs the other way: when a human cuts into an LLM's smooth, in-frame elaboration to ask "wait — why are we optimising this at all?", the human, entirely from inside their own frame, has just acted on the LLM's blind side. (Difference is necessary here, not sufficient: two frames could differ and still leave each other's blind spots dark. What the design leans on is the narrower fact that, in the cases that matter, one frame lights up what the other cannot — and §6's institution exists to *find and keep* those crossings, not to assume they always occur.)

Two everyday cases make the relationship concrete. *The human as the LLM's partial outside:* asked to raise a function's test coverage, an LLM will keep generating ever-more-thorough tests — "raise coverage" is the frame it was handed. A human cuts in: "coverage is not the point; we keep shipping the same three production bugs — test *those* paths." Entirely from inside their own frame, the human has lit up a corner the LLM was working blind to. *The LLM as the human's partial outside:* a seasoned engineer reaches for a database because every system they have built has used one; the LLM, carrying no such professional habit, asks whether the data needs to persist at all or can be recomputed — an option the expert's training had quietly filtered out.

An everyday version of the same shape: two drivers, each with a blind spot in their own mirrors, can still keep each other safe on a merge — not because either has a god's-eye view of the road, but because what is invisible to one is plainly visible to the other. Neither has to leave their seat.

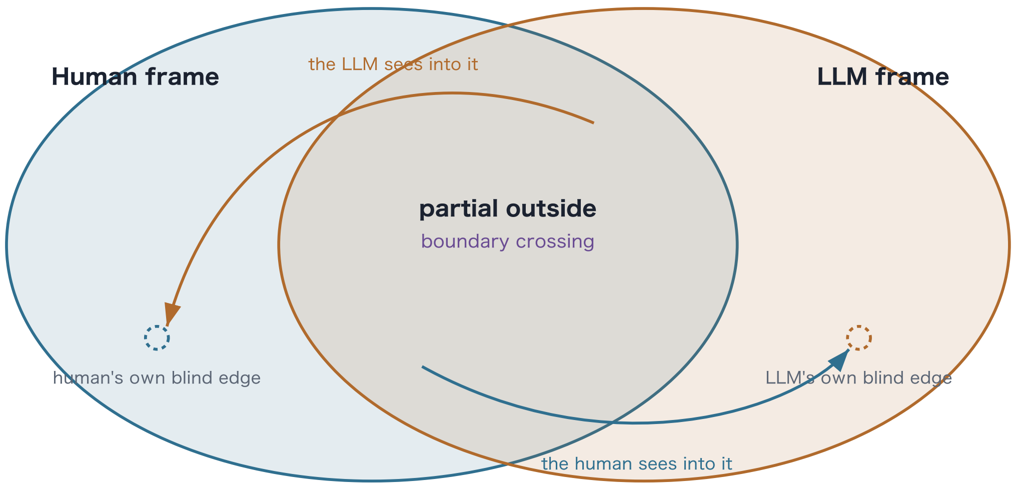
And the cases need not stay invented: the making of this very note recorded real ones, in the working-context layer of the system §§4–6 describe. One crossing ran from the models to the author. An early draft of §8 specified the self-embedding naively — hash the manuscript, record the hash on the chain, write the resulting record identifier back into the manuscript. In the first cross-model review round (2026-06-10), three reviewer personas independently converged on the same finding: writing the identifier in changes the bytes, so the deposited document could never verify against its own hash. The fixpoint was invisible from inside the author's frame and obvious from a verifier's; §8's present shape — a single frozen file hashed over its raw bytes, carrying no record identifier of itself, with the reference running one way from record to document — is that crossing's residue. Another ran the other way. A model-drafted version of the bridge that opens §5 asserted that a recorded insight, once *become*, "is reality" for the system; the author's line-by-line re-read (recorded 2026-07-12) caught the over-claim, and §5's present honesty — becoming puts an insight in force, it does not make it true — is *that* crossing's residue. Neither party transcended anything: both crossings are ordinary in-frame acts that happened to land on the other frame's blind side. What is less ordinary is that both are kept, datable, in the system's working record (kept, to be exact, in its *selective* tier — a distinction §5 will make much of), which is precisely the asset the design brief below asks for.

One asymmetry must be admitted right away, because §6's design leans on it. The symmetry of *frames* is not a symmetry of *power*. The human holds the agent's whole situation in their hands — they can end the session, rewrite the harness, decide what the agent is allowed to keep — while the agent has no comparable grip on the human. Both are frame-bound; but only one of the two frames sits inside the other party's control. This does not weaken the argument. It tells us where the boundary role has to live.

So the design brief for the rest of the note is set. If nobody can leave their own outermost frame, but each can reach the other's blind side, then the event worth caring about is the **boundary crossing** — the moment one party's content lands in a region the other could not have lit up alone — and the asset worth building is the **accumulated, inspectable record of those crossings**, because crossings only add up if they are kept. The next three sections describe a system built to host those crossings, record them, and let them compound.

Figure 2 — Partial outside

§2–§3 · the hinge of the argument



Neither leaves its seat · no god's-eye view · the frames differ, and that is enough

Nobody can step outside their own outermost frame. But **the two frames differ**, so what one keeps hidden from itself, the other sees plainly — the way two drivers cover each other's blind spots without either seeing the whole road. The event worth keeping is the **boundary crossing** in the overlap; difference, not transcendence, is what makes it possible.

4. Structural self-referentiality

The three sections from here to §6 turn that acceptance into design, through one worked example. **KairosChain** is the author's framework for agents that can change their own rules — one answer to what it would mean to take §1's Nomic seriously as software. (The argument so far is general; §9 states why it is nonetheless made on one concrete system.) Its generative principle is one sentence:

> **Meta-level operations are expressed in the same structure as base-level operations.**

Concretely: the things the system can *do* — its capabilities, called "Skills" — are written in a small Ruby-based language (a DSL) whose canonical form is the Ruby abstract syntax tree (AST). And the rules that govern how Skills are allowed to *change* — how one is promoted, rolled back, approved — are written in that same language, run by that same machinery, and recorded under the same discipline. "Defining a Skill" and "defining the rules for how Skills evolve" are not two different kinds of thing in two different places; they are the same kind of object. This is exactly Nomic's move structure carried into software: changing the rules is itself an operation performed *inside* the system, in the system's own language.

Two consequences follow. Each is deliberately a claim about *part* of the system, not the whole.

- **Partial autopoiesis.** "Autopoiesis" (Maturana & Varela 1980) means, literally, self-production — a system that produces and maintains the very components it is made of. The classic example is a living cell, which manufactures the enzymes and membranes that in turn manufacture it. KairosChain's self-production loop closes at the **governance level**: the parts that define its capabilities, and the parts that define how those definitions may change, are produced and maintained by the system's own operations. The loop does *not* close at the **execution level** — the system still runs on a Ruby interpreter, a filesystem, and an LLM that it did not make, much as a cell still depends on food and atoms it did not create. (Maturana and Varela themselves reserved “autopoiesis” for the living and would resist this extension; we borrow the vocabulary with that caveat noted.) The useful question is therefore not “is it autopoietic, yes or no?” but “*at which level does the self-production loop close?*” — and KairosChain answers deliberately: closed at governance, dependent at execution.

- **No transcendence claim.** Structural self-referentiality does not let the system step outside its own outermost frame. It does the opposite. It pulls as much of the system’s own constitution as possible *inside* the move structure, so that the line between “what the system can change about itself” and “what only its environment can change for it” becomes explicit, narrow, and open to inspection. The ceiling of §2 is not escaped; its location is made visible and put under management. Placing the authority in an outside controller does not reach past the frame either: as §2’s limiting case showed, a controller one level up only moves the ceiling one notch — it is a genuine outside of the *model* (a harness is exactly that), but not of the *frame as such*. What self-referentiality instead makes locatable is the seam where in-system expression runs out — the Ruby VM, the filesystem, the model, the human approval, none of them expressible as a Skill — and that seam is precisely what *partial autopoiesis* named just above: closed at governance, dependent at execution.

These two consequences are what §5 builds on. Pulling the constitution inside the move structure does more than make the ceiling visible — it settles what *kind of act* a governance change is. Carried out in the system’s own language and machinery, an amendment is the system’s own operation: something the system *becomes*, not something an outside authority does *to* it. Administered instead from a control tower one level up, each change would arrive as information handed in from outside — *had*, not *become*. And because that operation runs under the same discipline as any base-level one — the same language, the same machinery, the same recording — a governance change needs no separate, bolted-on recording mechanism; it inherits the base level's. Structural self-referentiality is thus the **precondition** for the move §5 turns on: a change the system *becomes* by being recorded. What kind of recording that becoming requires is the next section's question.

5. Constitutive recording, by construction

Before drawing the distinction this section turns on, one bridge from §3 should be made explicit — it decides what this section has to deliver. §3 ended by saying that crossings only add up if they are kept. What a crossing delivers is an **insight**: the thing one frame could not light up from inside and the other did — "coverage is not the point; we keep shipping the same three bugs; test *those* paths," in §3's example. Suppose the system keeps that insight in the obvious way: as a stored note. A note binds nothing. Whether it is consulted on the next occasion is a fresh decision every time, and nothing obliges anyone to make it — so the insight can silently stop mattering, which is exactly the selectivity failure diagnosed later in this section. For the insight to actually govern behaviour, storing it is not enough: the *system itself* has to change, so that the insight stops being a description sitting beside the rules and becomes part of the rules the system is bound by. Call the first state **having** an insight and the second **becoming** it; the rest of this section leans on that pair. How, then, is the change to be enforced? §4 has already closed the easy road: there is no outside control tower entitled to reach in and rewrite the system, so no external device can take the note and impose it. A governance change is only ever the system's own inside operation. That leaves exactly one place where an insight can be bound to behaviour — the operation by which the system changes itself

— and one way to make the binding unbreakable: make the act of recording and the act of changing **one and the same act**. Write the insight into a record whose entries *are* the system's rule-changes, and the gap between stored and in-force never opens, because a stored-but-not-in-force state no longer exists to fall into. One piece of honesty is owed at once: **becoming does not make an insight true.** Nothing can, from nowhere — that standpoint does not exist (§2). What becoming changes is the question that can be asked: not "has it been certified?", which nothing could answer, but "is it in force, and by what recorded path?" — which an audit can answer, and which a later, equally recorded repeal can correct. The device that performs this becoming is the constitutive record this section describes; and §4's same-structure discipline is what lets it sit at the governance layer — the layer that determines what the system is — with no second, bolted-on machinery.

The have/become distinction is one everybody already knows in the body. Knowing how to ride a bicycle cannot be *had* as a note: store every page ever written about balance and you still cannot ride; you can only become a rider — Polanyi's (1966) tacit knowing, with the cyclist as its classic case. The difference shows in how each is destroyed. A note can be discarded, because it is detachable from you; the riding cannot, because it is no longer information *about* you — losing it would make you a different person. The axis is therefore not implicit-versus-explicit but *detachable versus non-detachable* — and that is what licenses the twist this section turns on: an entry in the chain is an explicit, written, external mark, yet it sits on the *become* side, because a system stripped of its ledger is not the same system minus some records; it is a different system. The human's becoming happens tacitly; this system's becoming happens by explicit inscription. Opposite in form, identical in non-detachability. (The nearest theoretical relative of this "becoming" is the enactive tradition — Varela, Thompson & Rosch's (1991) *The Embodied Mind*, continuous with the autopoiesis vocabulary §4 borrows from

Maturana and Varela — for which knowing is never the possession of representations but the enacting of a world through a history of coupling. The kinship lies in what both deny: that knowledge is a detachable content its knower merely *has*. The departure lies in the means, and it is the same twist just noted — enaction works through a lived body; this system's becoming works through explicit inscription.)

Before turning to what a *record* is, one general feature of description itself is worth surfacing — with it in view, the evidential/constitutive distinction below reads differently. An action *in progress* is lived in what grammar calls the present progressive: the doing that is still underway, not yet closed. From inside that doing there is no completed fact to point at yet. What renders the action describable in the third person is its settling into the perfect — not "is happening" but "has happened" — and in an ongoing process the settling is no single terminus: it happens again and again, each confirmation precipitating one more completed fact while the doing runs on. Nor is the settling a neutral afterthought: what actually happened constrains what can truthfully be confirmed, but *which* completed fact the perfect fixes, within that constraint, is settled in the act of confirming, so the confirmation is part of how the deed acquires a determinate, third-party-approachable shape. This is the internalist reading of measurement Matsuno developed (Matsuno 1989), most explicitly in its tense-based form (Matsuno & Swenson 1999; Matsuno 2004): for an observer who cannot stand outside the system — the standing condition of §2 — description is not a mirror held up to a finished present but an internal act of completion, and a record registered in the perfect is precisely what turns a first-person doing into something a third person can approach (Matsuno 2013). One boundary keeps the borrowing honest. Matsuno's constitutivity concerns *describability*: every perfect-tense record — a ship's logbook included — takes part in settling what stands on record (not in making its content true). The distinction below runs on a different axis: whether *taking* *effect* passes through the record. The tense structure explains why a self-observing system's audit must reckon with records that do more than mirror; it does not, by itself, make any record constitutive in this section's sense. That further step is a coupling — and in this system it is engineered, not inherited from the tense structure.

With the have/become bridge in place, the second design commitment is about what a *record* is. Two very different meanings hide under that one word.

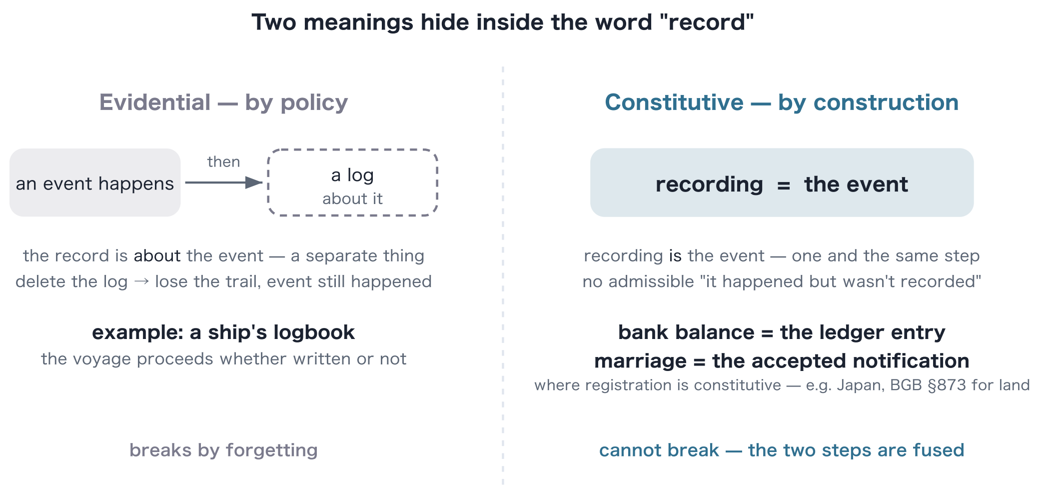
- **Evidential recording.** The system acts first; afterwards, a log writes down what happened. The record is *about* the system. Delete it and you lose your audit trail, but the system itself is unchanged — the events still happened. This is logging as almost everyone practises it, and where it is enforced at all, it is enforced *by policy*: a rule that says "remember to write it down," which some code path can always forget to honour. A ship's logbook is evidential: the voyage happens whether or not the captain writes it up.
- **Constitutive recording.** Here the act of recording *is* the act. A governance-level change takes effect **only by being recorded** — the entry appended to the hash-chained, append-only record is the change's very mode of existence, not a later description of it. There is no path by which a Skill is promoted, an evolution rule amended, or the instance's own constitution revised, that does not pass through the chain. (This is the *defining rule* of constitutive recording; stating the rule is not the same as having achieved it on every code path — a point we return to below.) Two familiar things already work this way. A bank balance is not a description of money kept somewhere else — the ledger entry *is* your balance; change the ledger and you have changed the money. In jurisdictions where registration is constitutive of marriage — Japan's, for instance — a legal marriage does not happen first and get recorded afterwards: the acceptance of the

signed notification into the register is what makes it. Under constitutive recording the system's governance identity, at any moment, simply *is* the irreversible sequence of recorded changes that produced it.

The contrast fits in one line: an ordinary log is a record of what the system *did*; a constitutive record is a record of what the system *is* — not because it describes a state, but because the inscribed sequence is itself the identity it would otherwise merely describe.

Figure 3 — The two meanings of "record"

§5 · evidential vs. constitutive



"Record" hides two meanings. In **evidential** recording the event happens anyway and a log describes it afterwards (a ship's logbook); the log can be forgotten. In **constitutive** recording the act of recording **is** the event (a bank balance **is** the ledger entry; where registration is constitutive, a marriage **is** the accepted notification); it cannot be forgotten, because recording and taking-effect are the same single step. KairosChain makes governance-level change constitutive — a change takes effect **only by being recorded**. (Honest limit, kept in the prose §5 / §9, deliberately left off this figure: this is a guarantee **by design**, and full outcome-independence — recording failed attempts and runs too — is not yet reached by construction anywhere.)

None of the constitutive register itself is claimed as new here, and its lineage should be named. It is Searle's (1995) account of **institutional facts** — facts sustained by collective recognition, for which declaration and registration are paradigm devices, and money and marriage his canonical examples, deliberately echoed above. The same line has long been drawn in law between *constitutive* and merely *declaratory* registration: under the German Civil Code, for instance, transfer of land ownership requires both the parties' agreement and entry in the land register, and takes effect only upon the entry (BGB §873) — the entry does not report the transfer; it completes it. At the level of software state, event-sourced architectures (Fowler 2005) make the event log the authoritative source from which all state is derived, and public blockchains carry the same idea to the letter — a bitcoin balance is nothing over and above its ledger entries. Searle supplies the social side of this lineage — declaration and registration constituting institutional facts; the descriptive side comes from Matsuno's internal measurement, met in the tense hinge above. The two are independent traditions, not halves of one account; they meet here because a self-amending system is at once an institution registering its own acts and an internal observer of them. What this note takes from that lineage — and, to our knowledge, is not stated within it, though adjacent literatures on on-chain governance and tamper-evident audit logging build machinery in the same space — is a narrower move: carrying the evidential/constitutive distinction into the **audit record of a self-amending system's own governance**, and locating the difference in the manner of enforcement, so that it becomes a property a system either has or lacks by design rather than a doctrine its operators must remember to follow.

The phrase **by construction** marks the difference between the two in one word: enforcement. A policy ("all changes must be logged") can be broken by forgetting. A construction cannot, because the recording step and the taking-effect step are, by design, *the same step* — there is no admissible version of “the change happened but wasn’t recorded,” just as there is no version of “the money moved but the ledger didn’t.” (How fully the running system realises this design — where a stray code path still separates the two — is the migration status §5’s later paragraphs report honestly.)

A natural objection makes the enforcement point concrete: *why not simply keep the Skills — governance rules included — in a Git repository?* Git already offers hash-linked history, signed commits, and review gates. The answer is not that Git is too weak, but that, used as a store, it sits on the wrong side of the distinction just drawn. Nothing in Git couples *taking effect* to *being committed*: the running system’s effective rules are whatever is on disk or deployed, and the repository holds what someone chose to commit — recording as discipline, the policy pole exactly. The governance act itself — who approved a change, and that the approval was required — lives in platform metadata (a pull-request record on a hosting service), outside the hash-linked history and inside a third party’s custody; and governance administered through an external platform is precisely the outside control tower §4 declines — the change arrives as something done *to* the system, had rather than become. None of this is a defect of Git, which was built to version content, not to constitute it. And §7’s concession applies here in reverse: a team that wired its runtime to treat the repository as the sole source of effective rules, recorded approvals inside the history, and made taking-effect and committing one step would not have refuted the design — they would have implemented it, with Git as the storage backend. The thesis is about the property, not the storage technology.

The register is now defined; what remains is the honest audit of it, in three steps — what even a constitutive record can omit (selectivity), exactly which half of an ideal it delivers (effectivity), and how far the running implementation has carried the design (migration). Three small events will keep the audit concrete. Suppose the system adopts rule A (event 1); a week later it quietly repeals A (event 2); and a proposal for rule B is voted down, changing nothing (event 3).

A third distinction cuts across the first two, and it is the one a sceptic reaches for fastest. Even a constitutive record can be *selective*. A system might make recording the mode of existence of the changes it does record, yet keep discretion over *which* acts it subjects to that discipline — attesting only the runs that flatter it, certifying only the results it is proud of. Selectivity hollows the guarantee from a different direction than forgetting does. A policy is broken by an *omission one was meant to prevent*; discretion is *omission by design* — and it is the more corrosive of the two, because where a policy leaves a gap you can at least point to, discretion leaves no gap at all: the absence of a record carries no information, since nothing ever obliged the record to exist. The ideal that closes this gap is **outcome-independence**: whether an act is recorded is not conditioned on how the act turns out, so that failures and dead ends are as present in the record as successes. In the language of the three events: outcome-independence would put event 3 on the record as surely as events 1 and 2.

Constitutive recording delivers a *part* of that ideal, and it is worth being exact about which part. A word of care about “effective” first, since the whole distinction turns on it: an act is *effective* here in the sense of *taking effect and being in force*, which — under the constitutive definition above — is the same as *being recorded*. Two readings therefore travel together and must not be silently swapped. In the **definitional** reading, “no effective change goes unrecorded” is true by the meaning of the terms; in the **behavioural** reading — the system actually did something — an act that ran but, through the implementation gap noted below, never reached the chain is a change that was behaviourally effective yet unrecorded. The design aims to collapse the two readings into one; the migration is how far that collapse has got.

What the construction delivers is events 1 and 2. Since recording is designed to *be* the act of taking effect, an effective governance change is meant to reach the chain at the very moment it is effective, before any downstream outcome is known. Adopting rule A is on the chain (event 1); so, crucially, is the repeal (event 2) — a **reversal that takes effect** is itself an effective change, so neither it, nor any promotion or amendment, can be silently omitted (the migration paragraph below marks where the running implementation still falls short of this by-design guarantee). That already defeats one species of cherry-picking: you cannot quietly roll a rule back and later present a history in which the rule was never there.

What constitutivity does *not* by itself deliver is event 3 — the other half of outcome-independence. A rejected proposal changes no state, so nothing obliges it onto the chain; and whether a given run of the system is recorded at all is, today, a matter of policy, not construction. Reaching those classes needs a recording gate that is not yet built by construction, and §9 marks it as open. So the honest reading is narrow, and still worth stating: at the governance boundary the record is non-discretionary over *what took effect* — events 1 and 2 are fact rather than curated evidence, and with them the *governance* trajectory: which rules were adopted, amended, reverted. The richer, results-level ideal — "record everything the compute did, win or lose" — remains an aspiration, not a standing guarantee.

KairosChain is designed around the constitutive register, and the concrete engineering work of its current development cycle is exactly the **migration** of its recording guarantees, layer by layer, from the policy kind to the construction kind. What matters for the argument here is the

governance layer — where Skills are promoted and the instance's own constitution is amended. There, recording is constitutive by design and closest to the construction ideal: a change is not meant to count as effective except through its chain entry, so the whole class of effective changes is non-discretionary. Even here the ideal is approached rather than finished: in the current implementation some changes are written to their files and only then recorded, rather than in one indivisible step — so a crash between the two steps, or a recording failure that is caught and merely logged while the change stays in force, can leave an effective change off the chain; closing both paths is the migration's current work. (The other layers carry deliberately different regimes — a selectively attested, opt-in tamper-evidence tier, and a disposable working-context tier meant to be discarded — treated where they belong rather than toured here: §6 places them in the L2→L1→L0 promotion path, and §9 draws the limit they mark.)

One scope limit belongs right here, not buried in a footnote. Words like "irreversible" and "append-only," for a chain run by a single operator on their own machine, are guarantees *relative to that operator's honest operation.* A hash chain makes tampering **evident** to anyone who already holds a trusted copy of the chain's latest hash; it does **not** make it impossible for the operator themselves to rewrite history and recompute every hash from the edit forward. What keeps the operator honest is **external anchoring** — committing the chain's state into some record the operator does not control — and §8 below is, among other things, exactly such an anchor.

Finally, constitutive recording changes the *kind of time* the system lives in. A log lives in **Chronos** — clock-time, one interchangeable instant after another, any of which could be overwritten without much loss of meaning. A constitutive record lives in **Kairos** — the Greek word for the charged, decisive moment that is not interchangeable with any other.

Within the trust scope just stated, each entry is a decisive, irreversible reconstitution of what the system is. The nearest philosophical relative is Whitehead's (1929) "objective immortality": a completed moment is not erased by the next but becomes a permanent condition that every later moment inherits. The reading of time this implies — that the irreversible sequence of records *generates* the system's time from within, rather than marking positions along a time given from without — is a governance-layer restatement of what Matsuno's internal measurement claims for material process at large: for an internal observer, time accrues through the repeated settling of doings into completed facts, not from an outside clock. (The external anchors of §8 live, deliberately, in outside time — that is what makes them anchors.) (Named plainly, this is the old divide between substance and process ontology: a log presupposes a system that exists first and leaves traces behind; a constitutive record makes the system the kind of entity whose being *is* its history of recorded events. The design takes the process side — at the governance layer, and only there: §4 already conceded that at the execution layer the system rests on substrates it did not make.) A boundary crossing from §3, once recorded, is then not information the system *has*. It is part of what the system *is*.

§4 and §5 are best read as a single move. §4 made the seam between what the system governs and what it merely runs on *explicit*; §5 made recording each crossing of that seam the very act by which it takes effect — *by construction* at the governance layer, so far as §5's migration has carried the guarantee. Where an ordinary system leaves this boundary implicit and unlogged, a self-amending one built this way makes it explicit and its crossings auditable from outside — not visible to the system from inside, which §2 forbids, but legible to an outside observer. The outermost-frame boundary, silent in an ordinary system, becomes a recorded one; that is the boundary §6 now builds a human into.

6. Institutionalising the human boundary

Sections 2 and 3 established two things: the agent cannot stand outside its own outermost frame, and the human — who cannot stand outside *their* own frame either — can nonetheless reach the agent's blind side, and moreover holds the agent's whole situation in hand (the power asymmetry of §3). §5 then secured the **form** of keeping: an effective governance change cannot stay off the record. What the recording machinery cannot generate is the **content** — *which* changes deserve to take effect. If the system judged that alone, the judgment would be handed down from inside its own frame — the blind spot grading itself — which the ceiling of §2 forbids. So that judgment needs the act of someone with a *different* frame, a partial outside (§3), built into it; and the deeper the layer a change touches, the heavier that involvement must weigh, until at the deepest step — amending the system's own constitution — it cannot be skipped at all. KairosChain turns this from a vague good intention into an institution, through three mechanisms.

- **The human at the boundary, not merely "in the loop."** The familiar vocabulary undersells what is required here. A *supervisor* watches and comments afterwards — ignorable, and it fails the way "forgetting to write" fails. Even a *gatekeeper* is still the evidential picture: someone standing outside the gate, inspecting what passes, able to doze off. What is built here is of a different kind. At the constitution-amending step, the human approval is a **component of the single step by which the change takes effect and is recorded** (§5): when the approval is absent, the gate is not closed — the gate *does not exist*, and there is no admissible state of "it took effect without the approval." This is constitutive recording extended to the human act. A bank's safe-deposit box carries the image: the door opens only when the clerk's key and the customer's key turn together. Compare that clerk with the guard at the camera monitors — the guard watches and can report afterwards, but he can doze off, and the door opens whether or not he is there; the two-key clerk is not watching the event but is *a component of it*. The human approval at the constitutional step is that second key. (At shallower layers a change can be recorded and take effect without human approval; the gate is constitutively built in at the constitutional step only — the next mechanism says which step that is.) The human, then, is not modelled as a supervisor watching from outside, nor as one more component bolted inside. The human stands *on the boundary*: their acts — approving a change, challenging it, reframing it — are the events through which the system's self-redefinitions actually pass. The system's boundary is partly *made of* those acts. The popular phrase "human-in-the-loop" undersells this; the loop in question is the loop by which the system defines itself, and the human is one of its load-bearing parts.

- **A promotion path that takes the outside in.** Consider what becomes of an outside insight — a reviewer's finding, or a reframing the model could never have reached from inside its own frame. In KairosChain it enters as throwaway working context (the L2 layer); if it proves its worth it is distilled into curated, reusable knowledge (L1); and it may eventually be written into the system's own constitution (L0). Each such promotion is recorded on the chain, and the step that writes into the constitution itself (the L1→L0 promotion) is human-gated. A concrete instance: a reviewer points out a blind spot in a design; that observation is first just a note in one session, then becomes a saved convention the system consults, and finally — if it earns it — a rule the system is bound by. This path is the standing machinery by which *the agent's blind side, once a human has externalised it, becomes part of the agent's own frame next time.* What the frame ceiling forbids as a single heroic act — selftranscendence — is not thereby *approached*: transcendence has no near side, no being-halfway-outside. It is **functionally substituted**: a repeated, recorded, two-party process delivers some of what a transcendent standpoint would have delivered, without anyone ever occupying one.

- **A harness that supplies what introspection cannot.** By the argument of §2, an LLM cannot reliably tell, just by looking inward, which layer of its own situation it is currently acting in — this is not a flaw to be trained away but a direct consequence of the ceiling: knowing it from the inside would require the very outside vantage §2 denies. So the harness *supplies that information from outside*: it injects, as explicit context, which layer is in play and which boundaries a given action would cross. The division of labour is principled. The model brings in-frame competence; the harness brings knowledge of where the frame's edges are; the human brings the partial outside.

Two things about what this institution brings into being deserve exactness. First, what comes into being is neither an enlarged human frame nor an enlarged model frame — neither party leaves its own (§3). It is a **third thing**: the composite system of human, model and record, with an outermost frame of its own. The frames have not fused into a higher ceiling; one more inside has come into existence, and that inside has a ceiling of its own — the regress §9 places last, and reads as an engine rather than a flaw. Second, the boundary work runs both ways. When a crossing lands in the *human's* blind spot and is taken in, what changes is not the human's stock of information but the shape of their seeing — the human's own side of §5's becoming, happening tacitly, as becoming in a body does. The chain holds the system's half of that joint history in a form a third party can inspect; the human's half it cannot hold at all — which is precisely the asymmetry §7 turns into a corollary.

None of this *solves* the ceiling of §2 — by the ceiling's own logic, nothing could. What it does is turn the ceiling from a silent failure mode into an explicit, instrumented boundary with recorded traffic moving across it. The stance is acceptance, not conquest — and acceptance, once engineered, turns out to be productive rather than resigned.

7. The economic corollary: experience as capital

§5 closed on an ontological point: under constitutive recording, what the system *is* is its history of recorded events. This section draws the economic face of that point. If the system's being is its recorded trajectory, then so is its **value** — and value of that kind is not the copyable mechanism but the trajectory itself. This is the sense in which the design turns accumulated **experience into capital**: not the machinery, and not even the record in isolation, but the anchored trajectory of boundary crossings (§3) is the asset. Its nearest economic cousin is **goodwill** — the intangible premium a going concern carries beyond its tangible parts — with one difference that is the whole point: goodwill is asserted and estimated, whereas a constitutive trajectory is *audited*.

The claim must be stated carefully, because a hash-chained record is, like any file, copyable bit for bit. What the construction guarantees is not that the trajectory cannot be *copied* but that a copy cannot be a *continuation*. Someone who copies the mechanism and the chain together does not get a clone of you; they get a **fork**. At the instant of copying, the identity thesis of §5 grants the copy the very same constitution — the bits are identical, so nothing distinguishes them yet. Originality is therefore not a property of the bytes; it is a property of the *subsequent anchored trajectory*. The moment each side begins recording its own boundary crossings, the two constitutions diverge, and the question “which one is the original?” becomes **a matter of audit, not of assertion** — settled not by the chain alone, which looks identical for either copy, but by the external anchors of §5 that pin one trajectory’s history to a public identity at public times (the DOIs and timestamps of §8, each anchoring the chain prefix up to its moment), and by the operator’s own ongoing work. The chain makes each trajectory attributable and tamper-evident; the anchors tie one of them to a public identity; together they supply *provenance*.

The governance-boundary non-discretion of §5 gives that provenance a particular kind of strength — but only as much as it actually earns, and the sharpest form of the worry is worth meeting exactly: *not “won’t someone copy it?” but “won’t someone fabricate a rival that looks just as good?”* Take the honest case first. An original that ran its chain honestly could not, along the way, quietly drop the amendments and reversals that actually took effect — within the governance layer those are, by design, non-discretionary — so its recorded *governance* trajectory is not a *retroactively* curated reel: whatever rules it adopted and later undid are both on the chain and cannot be edited out after the fact. (Curation at the proposal stage remains available — an operator can simply never let an embarrassing change take effect — so the guarantee is that the *effective* record is un-erasable, not that everything awkward was ever attempted.) That is a narrower asset than “every misstep ever made” — the results-level record of dead ends is exactly the selective, disposable material §9 leaves open — but it is a real one, and it is a part a rival cannot simply assert away.

The dishonest case is where the force actually comes from, and it does *not* come from the record being outcome-independent: a single operator can, as §5 concedes, rewrite a local chain and recompute its hashes, so nothing internal to the record makes failures unforgeable. The force comes from **external anchoring** — with two qualifications that keep the claim from over- or under-reaching. First, an anchor is punctate: the §8 deposit pins the prefix *up to its moment*, so the economically valuable stretch — the experience accumulated *after* the deposit — is only as anchored as the operator's subsequent deposits make it. Durable provenance is a cadence of anchors, not a single act. Second, and this is where the constitutive design earns what a bare timestamp cannot: an arXiv posting, a signed git tag, or a notary's stamp all anchor a *snapshot* — they certify that *these particular bytes existed at this time*, and no more, since one can always have kept other rules or run other code off to the side of what was committed. A constitutive governance chain, once anchored, certifies something stronger, because by design there is no effective governance change except through the chain: the anchored prefix attests not merely *some bytes at a time* but, on the recorded-governance class, *the effective-governance identity complete up to that time* — leaving a rival no recorded-and-in-force change to reveal later, and the original none to deny (complete over the recorded-governance class by §5's definition; over the *behavioural* class only so far as the migration has shut the write-then-record gap). That completeness is what a generic timestamp cannot reproduce. The premium is not, strictly, unique to a hash chain — a discipline that deployed *only* what it had committed would approach the same record↔effect closure; what the constitutive design contributes is to make that coupling **structural** rather than a procedure one must remember to follow. The defensible asset, then, is not secrecy and not the chain in isolation; it is a *complete*, non-retroactively-curatable governance trajectory, pinned to public anchors a competitor cannot forge after the fact.

This is exactly the situation of a forked code repository — a complete copy of a project's entire change-history, taken so that development can continue independently. Forking copies the entire history perfectly; what it cannot copy is which fork the community, the maintainers, and the signed releases go on recognising as the line of descent. The bits are shared; the lineage is attributed. (Copying only the mechanism, without the chain, is starker still: you get a system whose constitution is empty — a zero-history newborn in borrowed clothes.)

And the other half of the asset cannot be transferred at all, for entirely ordinary reasons. The **operator's** own transformation — the years of reframings a human gave and received, the human side of the same trajectory — is tacit experience in Polanyi's (1966) sense, knowledge that lives in the knower rather than in any detachable record, and no recording technology has ever copied tacit experience. That was always true of any expert. What constitutive recording adds is the *system's* half: it makes the system's side of the shared history inspectable, attributable, and divergence-evident, so that the pair's accumulated experience together is not merely real but *demonstrable* to a third party.

This is also the answer to the practical worry the design most often provokes — *if you publish the mechanism, don't you give away the advantage?* Under evidential recording you might: there the mechanism is the asset, so a copy of the mechanism is a copy of the asset. Under constitutive recording the value never lived in the mechanism, so the note can be published without undercutting itself — indeed, publishing is the move *consistent* with its own thesis. If the value lived in keeping the mechanism secret, "constitutive recording" would be false advertising. Because the value lives in the recorded trajectory and its human counterpart, being open about the mechanism costs nothing the design had not already declared uncopyable (the operator's tacit half) or fork-evident (the system's recorded half). In a future where the raw capability of frontier models is a commodity anyone can rent, the thing that distinguishes one human–agent pair from another is exactly this: the recorded history of their work together at the boundary. The design takes an old truth about good teams — that their real value is shared history, not the org chart — generalises it to human–AI pairs, and then makes that shared history tamper-evident and inspectable, to the degree §5's migration has carried the guarantee into construction.

8. Self-embedding: this document as its own demonstration

A note arguing that recording is *constitutive* ought not to settle for being merely *evidential* about itself. So this document is to be written into the very system it describes — not as an outside note *about* a contribution, but as an **L1 reference Skill entered into the instance's own knowledge and recorded on its chain**, so that the document becomes part of what the instance *is*. The procedure is specified here and carried out at the deposited version.

The deposited artifact is a single **frozen file**: the English (authoritative) Markdown source, with this working editorial-status note absent (it is scaffolding, not part of what is deposited), line endings normalised to LF and text to Unicode NFC, and nothing else. What is bound is the **SHA-256 of that file's raw bytes** — there is no canonical-form surgery and no field to blank. The rendered PDF and the Japanese translation deposited alongside are convenience copies; the English file's bytes are what govern in case of any discrepancy.

The document embeds only handles fixed *before* it is hashed, so that embedding them cannot move the hash. There are three, all content-independent:

- the reserved Zenodo **version DOI**;
- a **direct link to this note's source file** in the repository (a tag or branch path — not a commit-SHA permalink, which would depend on the content);

- a **verification address** — a stable handle under which the record that anchors this file is registered.

The steps follow.

1. **Freeze and hash.** Produce the frozen deposit file with the three handles embedded, and take the SHA-256 of its raw bytes.
2. **Commit constitutively.** Record that hash on the originating KairosChain instance's chain — the constitutive record by which the document, held as an L1 reference Skill, enters what the instance *is*. The record binds the **verification address** to the file's hash, alongside the DOI and repository references. The document is not handed a record identifier in return; it carries none.
3. **Publish and demonstrate.** Deposit the frozen file on Zenodo under the reserved DOI — the **public reference**, and, being a record the operator does not control, the **external anchor** of §5, pinning the chain prefix up to and including the committed record. On the shared Meeting Place, register the matching **attestation** — the digest bound to the verification address, pointing at the Zenodo and repository copies. The body lives on Zenodo; the Meeting Place holds only this reference record, which is the **site where the constitutive recording is demonstrated**, entering the document into the federation. For this operator's own chain that attestation is a public reference under the same operator's control, not a further external anchor; for an instance under a *different* operator the same attestation is itself a genuine external anchor — so the anchor's role is relational (§5).

Why does this not chase its own tail? The worry looks real — the document is recorded, and the record refers to the document — but the reference runs **one way only**, from the record to the document, never from the document to a fingerprint of itself. The document carries no hash of itself and no record identifier, only the three content-independent handles fixed before hashing, so there is nothing self-referential in its bytes to resolve. A third party downloads the file from Zenodo, takes its SHA-256, and compares it to the hash held in the attestation that the verification address resolves to — recomputing the fingerprint independently, which is what makes the check meaningful rather than circular. The loop that remains is the intended one, a strange loop in Hofstadter's exact sense: a document about constitutive recording, constitutively recorded — its publication, the very act §7 argued was safe, becoming one of the recorded events that constitute the instance. That the loop lives in the *act of recording*, not in any number the document prints, is exactly why the document need carry nothing of itself.

We flag, with a smile rather than an apology, that this section does not escape §2 either: embedding the document in the chain is one more move inside the system's frame, not a step outside it. That is the point.

Nor does it escape §5's own distinction, and it is worth saying so plainly now that §5 has drawn it. What this embedding demonstrates is *constitutivity* — a single completed artifact, bound to the chain and anchored in public — not *outcome-independence*. The note's own making is the opposite pole: its drafting history, v0.1 through this version, with every reversal a multi-round review left behind, is exactly the selectively-curated, un-anchored material §5 assigns to the disposable layer. What gets deposited is the surviving draft, not the record of the drafts that did not survive. So the document that argues the distinction is, in its own construction, an instance of the *weaker* side of it — a constitutively-committed conclusion drawn from a selectively-remembered process. That is the honest place for it to stand, not a contradiction to be hidden; it is the same asymmetry the system itself lives under, performed once more by the note about the system.

9. Limitations and open questions

- **Scope.** This is a design note; it makes no new empirical claims. The outermost-frame ceiling (§2) is argued in the strange-loop lineage, not measured; the empirical study of *that* limit belongs to the companion note in preparation and, we hope, to independent replications using the public 2009 instrument. The design answer of §§4–6 is of a different kind: it is not proven on paper but meant to be *demonstrated in operation* — and our hope is that KairosChain's own continued running, where each effective governance change takes effect only through its record, is where this note's thesis is borne out rather than merely asserted.
- **The boundary role is not exempt.** §6 places a human at the boundary, but the human's acts are themselves frame-bound (the symmetry of §3). The design does not claim the human supplies a view from nowhere — only a view from *elsewhere*, backed by the embedding-power asymmetry §3 concedes. Whether parts of the boundary role can be played by other agents with sufficiently different frames (other model families, other instances with divergent recorded histories) is open, and the system's federation layer is built to test it.
- **Selective attestation is not outcome-independence — and neither, yet, is the governance layer.** Between the constitutive governance layer and the disposable working-context layer sits a **middle instrument**, recently built: an *opt-in* tamper-evidence tier for individual working-context units. A human picks a working note that deserves to survive scrutiny, and its content digest is fixed, at that moment, in an append-only, hash-linked proof store; a later correction receives a new certificate layered over the old one — **non-destructive supersession**, never erasure. This buys inspectability at

the price of discretion: a human decides *what* is attested, so the *absence* of an attestation there proves nothing, exactly the weakness §5 warns against — and, at the time of writing, this tier is not yet externally anchored, so its guarantee holds only within the honest-operation model of §5. That is the right trade at the throwaway-context layer, where selecting *against* stored context is the whole point. But the deeper limitation is that full outcome-independence — recording attempts that failed and runs whatever their result — is not delivered *anywhere* by construction today, not even at the governance boundary, which is designed to secure only the narrower thing — that effective changes cannot be hidden — and secures even that to the degree the migration of §5 has closed. Where the line should fall — which classes of act must be recorded regardless of outcome, and which may be left to human selection — is a live design question, not a settled one, and it is the same boundary problem any run-time recording gate must eventually adjudicate: a meaningful negative result deserves the outcome-independent regime, while pure noise (a syntax error, an aborted keystroke) belongs to the disposable one, and the line between them is not yet drawn by construction.

- **Regress, expected.** Any institution that takes the outside in creates a new inside, whose outside must again be supplied from without. We read this not as a flaw but as the engine: complete self-description being unavailable in principle, the system's evolution is *driven* by the permanent residue of what it cannot yet describe about itself. The promotion path does not end the regress; it harnesses it. This reading — that the unresolvable residue is what drives the system rather than a defect to be closed — is where the internal-measurement lineage (Matsuno) arrives from the side of natural description: for an observer with no outside, the incompleteness is not removable, but productive.

- **Generality.** The argument is stated for one system (KairosChain) because constitutive recording is a property a system either has by construction or lacks; it cannot be claimed in the abstract. The design pattern — same-structure self-reference, construction-level recording, institutionalised boundary — is offered for reuse, with the corollary of §7 as the reason reuse does not dilute the original.

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